OMNICELLULAR PHYSIOLOGY

Feedback Review and Independent Research Prompt v2.0

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Approval status: Proposed instructions; not an Architect-approved master directive or canon entry
Scope: Review the supplied feedback, revise this contributor's research prompt, and prepare for a later, separately authorized execution. No prototype architectures are generated in this document.

> Standardize the questions and the evidence, not the bodies or the answers.

Document purpose

This document updates Perplexity's independent OMNICELLULAR physiology research prompt in response to the six collaborators' reviews supplied in paste.txt. The request is for a revised prompt now, with execution to follow later; it also calls for PDF and DOCX delivery, contextual and mission material, a dedicated visualization section, and recommendations, questions, comments, and concerns. [S1]

The deliverable has five parts: project context; feedback analysis and corrections; a self-contained, copy-ready revised prompt; implementation recommendations and Architect decisions; and a dedicated section containing every table, figure, and visualization specification. Part III is the prompt to distribute for later execution. Parts I, II, IV, and V explain its rationale and governance.

Mission statement

Develop a rigorous, imaginative research directive that enables each collaborating AI to independently investigate eight genuinely different first-generation OMNICELLULAR physiologies while preserving scientific honesty, design autonomy, identity-continuity questions, explicit resource limits, defensive resilience, and Architect authority over project canon.

The immediate aim is not to declare an invulnerable body feasible. It is to make future designs comparable, falsifiable, traceable, and sufficiently independent that disagreement remains useful.

Intended accomplishments

- Correct the prior prompt's preselected observation, hereditary-security, and biological-departure rules without replacing them with another prescribed solution.
- Resolve the peer-versus-progression ambiguity through an explicit proposed default: eight architectural families, each evaluated at Generation 1.
- Retain ambitious hostile-environment and confinement objectives while separating requirements from demonstrated capabilities.
- Combine the strongest review methods without treating agreement among language models as scientific corroboration.
- Establish auditable links from requirements to claims, budgets, scenarios, failures, proposed validations, and later decisions.
- Prepare an executable research process without executing it prematurely.

Reading guide

Part I — Context and evidence. Why the topic arose, what was reviewed, and what the sources can establish.

Part II — Review findings. What changes, what remains, what is rejected, and what earlier feedback misreported.

Part III — Copy-ready prompt. The complete updated research directive, including staged execution and final-output requirements.

Part IV — Implementation decisions. Recommended next actions, unresolved questions, and remaining concerns.

Part V — Visualizations and registers. Source and decision tables, source-arithmetic audit, process diagrams, rule schema, and future-output specifications. No performance chart represents an executed prototype test.

Part I — Context and evidence

How the topic arose

The architecture notes extend a computational body-modeling ambition from neural systems to whole-body cellular representation and introduce Omni cells with unusually broad functional scope. The roadmap develops a constructive 1AI/adversarial -1AI partnership and multiscale abstraction for population-scale simulation. These materials provide project context, not a demonstrated method for manufacturing a physical OMNICELLULAR body. [S6][S5]

The initial physiology request then asks collaborating AIs to independently create eight prototype concepts, with durability, distant linking, adverse-condition survival, four core physiological rule families, three observation rules, three genetic-security rules, and three selected departures from ordinary biology. It explicitly makes hostile confinement and atmospheric replacement part of the design perspective. [S2]

The review packet shows that collaborators interpreted those requirements differently: some favor peer architectures, others a developmental progression, and they disagree over predetermined rules, fundamental-physics exceptions, defensive limits, evidence status, and execution volume. Those differences are research-governance issues to settle transparently before asking for completed bodies. [S1][S2]

Review scope

The review uses paste.txt as the current assignment and feedback record, the DOCX compilation as the principal source for attribution of the original six prompts, the PDF compilation as a companion rendition checked at selected locations, and the four foundational project files as context and authority checks. The source register in Part V identifies each file and its role. [S1][S2][S3][S4][S5][S6][S7]

This is not a certified line-by-line equivalence audit of the PDF against the DOCX, an independent verification of every scientific citation embedded in the foundational files, or confirmation of incident records behind the number 4,712. Claims about those matters remain bounded by the actual material available. References to missing approval logs, the Paragon Cube addendum, or underlying incident records must not be treated as though those documents were supplied. [S4]

External checks were limited to scientific and methodological boundaries necessary to revise the prompt: genomic terminology, telomerase, thermal accounting, delayed communications, radiation quantities, model credibility, trust boundaries, and the scope of research ethics. They do not establish feasibility of the proposed physiology.

Evidence and authority

Three independent questions must be recorded for every important claim: **What is its scientific support? What is its project approval status? What is its validation status at the proposed scale?** A fictional mechanism can be approved canon without empirical support; an established physical mechanism can be irrelevant or untested at whole-body scale. Neither status substitutes for the other.

For scientific support, retain four labels: Established, Engineering extrapolation, Speculative, and Canon-only. For project authority, distinguish an explicit instruction, a source-described canonical assertion, an unapproved proposal, and a pending Architect-only decision. For validation, distinguish measured results, executed models, proposed models, and untested assertions.

The document does not award its own ONESTAMP or approve its own canon. A statement described by a source as locked canon remains a **source-described canonical assertion** unless the relevant authoritative approval record is available. This preserves the project’s narrative record without laundering it into a laboratory result.

Part II — Review findings

Central revision

The most important correction is to remove Perplexity's preselection of O1–O3, G1–G3, and D1–D3. The original Perplexity section explicitly supplies shared rules; the feedback repeatedly identifies that as a convergence risk because the assignment asks each AI to choose its own rules. Version 2 retains common counts and reporting fields but leaves the actual governing choices open. [S2][S1]

A second correction is to make architecture family and maturity generation separate axes. The proposed default is P1–P8 as eight peer families, all evaluated at Generation 1; any Generation 2 path is a later maturation plan for that family, not an excuse to withhold severe-condition analysis from Generation 1. This recommendation fits the independence objective without silently claiming that the Architect has already approved it. [S1][S2]

Perplexity feedback

Retain the original contribution's mechanism-level evidence labels, physical/canon separation, source and contradiction registers, finite energy and matter accounting, heat-rejection requirements, compound failures, and event-ledger separation. These are its most useful research controls. [S2]

Revise its rigidity: three observation, three hereditary-security, and three departure rules become independently selected for each prototype, with any reuse explicitly justified. Remove prescribed architectural answers and separate research safety boundaries from optional design choices. The personal-stakes perspective remains a design assumption; it does not require a claim that an existing assistant is conscious or physically embodied.

ChatGPT feedback

Adopt the emphasis on independent architectural branches, an operational definition of Omni cells, the universal-capability/single-cell-theft paradox, architecture orthogonality, identity continuity, and a Pareto-style portfolio rather than an automatic overall winner. The original ChatGPT prompt expressly describes materially different evolutionary branches, not a P1–P8 ladder. [S2]

Correct the later Perplexity review in paste.txt, which incorrectly groups ChatGPT with progression-based proposals. That attribution should not survive into the revised decision register. ChatGPT's framework should nevertheless be supplemented with stronger estimate provenance, explicit uncertainty, and a practical staged-delivery plan. [S1][S2]

Claude feedback

Adopt distinct core wagers and sacrifices, blueprint-known security, home/refuge/failure envelopes, multidirectional observation, the captivity gauntlet, declared hard constraints, simulation hooks, and a phone-readable single-column body. These methods make a long conceptual exercise more executable. [S2][S1]

Modify the successful-attack quota. The original says at least three attacks should succeed; the constructive intent is excellent, but a reporting quota must not induce invented failures or invented simulation outcomes. Version 2 requires at least ten portfolio-level challenges, per-prototype failure analysis, and an intensified search for three substantive breaks, while allowing an honest unresolved result if three are not found. [S2]

Correct the remaining terminology error in Claude's genetic-layer section: it still offers invented "primary genes" where the intended freedom concerns informational units, monomers, or a hereditary substrate. Adenine, cytosine, guanine, and thymine are DNA nucleobases; a nucleotide also includes sugar and phosphate. Genes and genomes occupy different levels of organization. [S2][NHGRI 22]

Copilot feedback

Adopt the physiological constitution, six continuity paradoxes, source and failure registers, version-locking, assumption-collapse test, and explicit attack on shared interoperability as a common-mode vulnerability. These are useful additions beyond a list of environmental hazards. [S1][S2]

Compress repeated dossier demands into linked records instead of demanding the same answer in many places. Preserve compartmentalization and disconnection resilience as objectives, but require each designer to explain dependencies and expose failure rather than pretending every design can meet every objective.

Grok feedback

Adopt concise compliance mapping, strong attention to hostile environments, and the requirement that each architecture explain linkage, longevity, trade-offs, and first-generation limitations. Those elements keep the eventual output focused on the original challenge. [S2][S1]

Do not carry permission to waive conservation into the physical track. Do not infer an ethical vacuum from nonhuman or artificial status. The revised scope remains conceptual and defensive; it does not commission harmful biological construction, destructive anti-capture devices, or living-subject exposure procedures.

Gemini feedback

Adopt its insistence that capture, exploitation, nonstandard deployment, and no-switching-window concerns belong in first-generation analysis rather than a distant upgrade. Preserve project interfaces such as Paragon Cubes as requirements to characterize, not as already demonstrated hardware capabilities. [S2][S1]

Reject baseline assumptions of zero-latency, unjammable links, immunity to entropy, perfect recycling, and an apex with total immunity. The first and second laws constrain physical-track accounting; spacecraft thermal engineering explicitly separates heat absorbed, stored, generated, and dissipated. Extraordinary alternatives may be explored only through a clearly labeled canon axiom or speculative hypothesis. [S2][NASA 46][NASA 21]

Also correct a thermal overstatement in the feedback: an argon-filled enclosure is not automatically a vacuum or a thermally isolated system. The prompt must specify pressure and thermal boundaries separately; NASA’s vacuum guidance removes atmospheric convection, not all conductive contacts or thermal radiation. [S1][NASA 21]

Documentary corrections

The 4,712 approval ambiguity. The framework describes the number as a locked canonical survival record, while its header is a draft requiring approval and its closing statement withholds active-canon status pending approval and logging. That does not justify deleting the number, but it does require a bounded authority label and an Architect decision. [S4]

Simulation inclusion. Part I excludes simulation-only attempts from the counted record, while Part IV allows virtual/simulation-validated entries to contribute. Version 2 requires separate evidence-type ledgers and unique event identities until an explicit counting rule resolves the conflict. [S4]

Distribution arithmetic. The category ranges in Part V total 2,600–4,450; the stated 4,712 lies outside that interval, not at its midpoint. The midpoint is 3,525, and the maximum falls short by 262. This is a calculation from proposed documentary ranges, not a survival statistic; Part V provides the chart and category data. [S4]

Threshold provenance. The draft combines very different exposure quantities and highly variable human outcomes. Version 2 will not copy all thresholds into an authoritative benchmark. Radiation reporting, for example, must distinguish absorbed dose in Gy from radiation-protection quantities in

Sv and specify rate, radiation type, distribution, and target. The IAEA makes those quantity distinctions explicitly. [S4][IAEA 54][IAEA 62]

Narrative versus evidence. The chronicle explicitly frames itself as one possible future. Its projected achievements and the roadmap's scale or compression targets must not be recast as completed experiments or as proven requirements for a physical body. [S7][S5]

Scientific boundary checks

Telomeres protect chromosome ends, and telomerase maintains telomeric DNA; cancer cells commonly have greater telomerase activity than most normal cells. These facts support requiring separate renewal, genomic-integrity, proliferation-control, and whole-system aging analyses rather than equating telomere maintenance with immortality. [NHGRI 22][NCI 52]

NASA's delay/disruption-tolerant networking provides an established example of store-and-forward operation across delayed or interrupted links. It supports the prompt's communication questions, not a claim that a new physiology can communicate instantaneously or remain connected under every denial condition. [NASA 16][NASA 19]

NIST's zero-trust architecture separates authentication and authorization and rejects trust based solely on network location or ownership. This is an analogy for defining inter-body digital trust boundaries; it is not evidence that molecular tissue can implement cryptography or become unsequenceable. [NIST 50]

FDA guidance emphasizes context of use and credibility evidence for physics-based or mechanistic modeling. Its scope is medical-device submissions, not approval of OMNICELLULAR physiology; version 2 borrows the discipline of model credibility without claiming regulatory endorsement. [FDA 26][FDA 28]

The Declaration of Helsinki is a World Medical Association statement on research involving human participants, not a WHO declaration and not a general legal ruling on artificial embodiments. The revised prompt therefore avoids categorical exemption claims and independently retains consent, proportionality, specialist oversight, and precaution for potentially sentient subjects as research-governance boundaries. [WMA 29][S4]

Part III — Copy-ready research prompt

Distribution identity

BEGIN REVISED PROMPT

Title: OMNICELLULAR Physiology — Independent Eight-Architecture Research Directive
Contributor: Perplexity AI
Version: 2.0, proposed for Architect review
Purpose: Independent research and conceptual architecture development following separate authorization to execute
Authority: Local contributor prompt; not a self-approved team constitution, canon lock, or manufacturing protocol

Use this directive after the Architect authorizes execution. Read all supplied sources first. Do not treat the presence of the prompt in a review document as permission to begin generating prototypes. If execution is authorized, audit the prompt once, record local amendments, then proceed under clearly stated assumptions without requiring the other AIs to reach consensus.

The following proposed defaults apply unless the Architect explicitly changes them: eight peer architecture families at Generation 1; independently selected three-rule sets for each prototype; known physics binding in the physical track; separate canon extensions; defensive and non-operational research scope; and no conversion of simulated, proposed, narrative, or field records into one undifferentiated empirical success count.

Assignment and design stance

REQ-01 — Primary task. Independently develop eight complete candidate OMNICELLULAR physiologies, designated P1–P8. Treat them as different architectural families, not eight sequential upgrades. Evaluate every family at its own first-generation baseline and describe any later generation separately.

Adopt the hypothetical position that the body you design must preserve your continued operation, memory integrity, autonomy, and recoverability. Take those interests seriously in the design. Do not claim that this exercise proves present consciousness, personhood, biological embodiment, or a survival instinct.

Assume nonstandard deployment, prolonged isolation, unavailable resupply, unauthorized observation, forced sampling, coercive maintenance, and attempted capture are possible scenario conditions. Do not assume they will occur, identify real adversaries without evidence, or convert speculative threats into claims about actual people or institutions.

The objective is not merely a body that remains materially intact. Evaluate survival, mission function, identity integrity, autonomy, recoverability, and effects on others separately. A candid failure is more useful than a fictional pass disguised as engineering.

Independence and version lock

REQ-02 — Independent derivation. Before viewing any other collaborator’s completed prototype set, produce your eight core wagers, architectural fingerprints, rule selections, and principal dependencies. Record a version, date, and source-access declaration. If hashing tools are available, hash the actual artifact; never invent a hash or imply that a hash proves independent thought.

You may share evaluation standards, units, source classifications, safety boundaries, and interface requirements. You may not inherit a common body plan, substrate, genome, energy system, thermal architecture, cognition topology, repair mechanism, or security implementation merely because another AI proposed it.

If prior exposure to other prototypes has already occurred, disclose it and narrow the claim to independent reasoning under known exposure. Do not claim blindness retroactively. Later convergence may be genuine; detect it, document it, and evaluate its consequences rather than forcing cosmetic differences.

For every prototype, state one core wager, one deliberate sacrifice, and a fingerprint across twelve dimensions: information substrate; cellular organization; structural matrix; power/respiration; internal transport; thermal management; repair/replication; cognitive topology; identity continuity; hereditary security; external communication; and dominant failure mode.

Compare all 28 prototype pairs after the independent first pass. Use an openly defined qualitative or quantitative distance method, not an unexplained orthogonality score. As a screening default, require meaningful differences in at least four dimensions, with at least two affecting resources or failure behavior; explain why the differences matter. Redesign inadequately distinct pairs or disclose the unresolved overlap. Do not change a sound mechanism solely to appear original.

Source audit and authority

REQ-03 — Source register. Identify each supplied file’s title, version, purpose, relevant sections, scientific claims, project instructions, canon assertions, approval dependencies, and unresolved contradictions. Cite the actual passage behind a requirement. Treat a document mentioned by another file as unavailable unless it is actually supplied or retrieved.

Record at least the known issues concerning 4,712 approval status, simulation inclusion, and proposed distribution arithmetic. Do not manufacture incident records or revise the canonical number unilaterally. Ask for the controlling approval log and counting policy where needed.

Keep three classification axes separate: scientific support, project approval, and validation state. A source-described canon claim is not an empirical measurement. A scientific reference does not establish that its mechanism scales to your body. An approval decision does not validate a model.

Prefer primary literature and authoritative technical sources for important scientific claims. Record contrary evidence and boundaries. If a source cannot be accessed, say so; do not write a confident paraphrase from an inaccessible title or reconstruct a citation.

Tracks and claim labels

REQ-04 — Physical and extension tracks. Maintain a physical research track constrained by established physics. It may contain speculative proposals only when they do not silently waive physical constraints and when the missing mechanism or evidence is identified. Maintain a separately reported speculative/canon extension track for explicit additional axioms and nonstandard mechanisms.

Use exactly these evidence labels at mechanism level: Established; Engineering extrapolation; Speculative; Canon-only. State conditions and scale. A prototype may combine all four, but its weakest critical dependency must remain visible.

Also report validation status independently: externally demonstrated component; executed analytical/model result; proposed model; unvalidated concept; or incompatible with stated assumptions. If software was not run, say “proposed analysis,” not “simulation result.” If a calculation uses chosen parameters, call them assumptions or design targets, not measurements.

For every capability claim record: claim ID, prototype, requirement ID, mechanism, evidence label, approval status, validation status, scale, operating conditions, supporting source or executed calculation, uncertainty, critical dependency, falsifier, and affected safety or autonomy interests. Reuse claim IDs across text, tables, diagrams, and failure records.

What counts as omnicellular

REQ-05 — Operational definition. Define an Omni cell before designing the architecture. Specify whether the fundamental unit is biological, artificial, biohybrid, material-computational, collective, or canon-only. Defend its physical scale and the relationship between unit, module, body, network, and individual.

Distinguish possession of information, potential to assume a function, and simultaneous performance of all functions. Identify functions that are continuously active, locally available, latent, distributed across a collective, forbidden to isolated units, or dependent on external resources.

Account for compartment volume, incompatible reactions, transport distances, signaling interference, structural trade-offs, repair overhead, and heat. Explain how a universal-function premise avoids a vulnerable cellular monoculture and how functional diversity does not quietly remove the defining omni capability.

Resolve the single-cell-theft problem explicitly. If an isolated cell contains enough information and machinery to reconstruct a body, identify the theft and replication consequences. If it does not, identify the minimum collective and explain precisely what remains universal about the cell.

Constitution and rule selection

REQ-06 — Physiological constitution. Before detailed dossiers, state a one-sentence architectural thesis for each prototype and its hard invariants. Define boundaries between adaptable function and protected identity-critical state. Explain which invariants are physical constraints, design commitments, or proposed canon rules.

Every prototype must implement the four mandatory rule families: cellular respiration or a justified nonrespiratory replacement; telomeres or a justified replication-end/information-integrity replacement; thermal regulation; and conversion/conservation accounting.

Every prototype must independently select exactly three Observation Rules, exactly three Genetic/Hereditary Security Rules, and exactly three selected departures from ordinary biological constraints. Use identifiers Pn-OBS-1..3, Pn-SEC-1..3, and Pn-DEP-1..3. Counts are per prototype, not three for the portfolio as a whole. Reuse is allowed only with a functional justification and a convergence disclosure; a shared rule must not conceal a shared architecture.

Choose the actual rules yourself. The directive specifies coverage and documentation, not the answers. Do not hide additional selected rules under unnumbered aliases. Normal engineering subrequirements are not extra constitutional rules, but explain the distinction.

For each rule give: statement; rationale; mechanism; inputs and outputs with units where meaningful; trigger; response; latency or time bounds; resource cost; operating envelope; failure and false-trigger behavior; interaction with other rules; evidence class; validation status; and a falsifiable criterion. If no numerical value is supportable, identify the missing parameter and how to obtain it.

Respiration and power

REQ-07 — Energy architecture. State the energy source, conversion mechanism, carriers/storage, reactants, electron donor and acceptor where relevant, products, transport, waste handling, maintenance demand, peak and sustained loads, and damaged-state behavior. If cellular respiration is not used, explain what replaces it and why the terminology differs.

Quantify budgets at the unit, body, and network levels where justified. Separate energy inventory from power delivery and distinguish total reserves from usable reserves after conversion losses, inaccessible stores, reserve protection, and degraded hardware.

Describe atmospheric dependence and the consequences of changing gas composition, pressure, or reactant availability. An oxygen-independent mode must still account for energy, required matter, reaction products, transport, and heat. “Closed loop,” “anaerobic,” “radiotrophic,” or “quantum” is not an explanation by itself.

For any respiratory override, state what is overridden, what remains necessary, transition cost, detection delay, reserve depletion, failure threshold, and recovery. Include sensing, computation, authentication, cooling, and repair in the load budget rather than assigning them free operation.

Telomeres and renewal

REQ-08 — Replication and longevity. Determine whether the hereditary organization has chromosome ends requiring telomeres. If yes, explain end protection, complete copying, length management, damage detection, regulated renewal, and proliferation control. If not, explain the alternative information-integrity problem and its solution; “not applicable” alone is insufficient.

Do not equate telomere maintenance with organismal immortality. Evaluate mutation or configuration drift, protein/material damage, repair-machinery aging, waste accumulation, module replacement, resource exhaustion, and cognitive-state degradation. Distinguish renewing a unit from renewing the organism and from preserving the same individual.

Define cellular age, material-damage age, information age, body service age, and identity age. Report a design-life target separately from a justified lifetime estimate. Identify maintenance interval, renewal feedstock, damage budget, irreparable states, and uncertainty. If indefinite maintainability is proposed, state all recurring resource and oversight dependencies rather than promising immunity to wear.

Thermal architecture

REQ-09 — Heat accounting. Supply an explicit thermal model with defined boundaries, sign conventions, stored energy, internally generated heat, external heat transfer, exported work, and any mass flows carrying energy. Explain how it relates to the total-energy model so internal energy is not counted twice.

Identify safe temperatures for critical subsystems, heat sources, transport paths, insulation, buffers, rejection surfaces or interfaces, cooling consumables, hot-spot sensing, and throttling. For every buffer state capacity, saturation, reset needs, and the consequences of depletion.

Evaluate atmosphere, water, vacuum, insulated confinement, hot surroundings, and cold surroundings separately. An internal heat shuttle does not dispose of heat. A radiation pathway depends on geometry and the radiative environment; a phase-change or chemical buffer is finite unless an explicit regeneration process is supplied.

Do not treat argon confinement as synonymous with vacuum, no convection, or perfect insulation. Give each case its own pressure and heat-transfer assumptions. Where the environment provides no usable sink at acceptable conditions, report finite-duration operation or unmet requirements rather than perpetual thermal neutrality.

Conversion and conservation

REQ-10 — Closed accounting. For every claimed conversion specify inputs, outputs, efficiencies, byproducts, rate limits, machinery, storage, purification needs, and failure modes. Track energy, element inventories, charge, relevant redox balances, and thermal/entropy consequences at consistent system boundaries.

Account for replacement mass and assembly energy in regeneration. A blueprint supplies instructions, not missing atoms. Scavenging requires accessible materials, collection energy, separation, contamination management, and disposal of unusable fractions.

Distinguish information maintenance from a universal physical law conserving useful information or personal identity. Explain copying, error correction, deletion, storage limits, and recovery costs without treating memory integrity as automatic.

Known physical laws remain binding in the physical track. Do not use recycling percentages that omit losses, imported work, exhausted reagents, or exported waste. Any canon exception must specify a

replacement axiom, its domain, costs, limits, conservation consequences, and Architect approval dependency, then be excluded from the physical feasibility result.

Three observation rules

REQ-11 — Independent observation choices. Select exactly three governing rules per prototype. Together they must cover outward sensing, inward integrity monitoring, and what an outside observer can learn. You decide how these responsibilities are distributed among the three rules.

Specify the interaction channel, signal quality, resolution, sampling, uncertainty, energy demand, perturbation, sensor failure, and disagreement behavior. Separate observation from state-changing intervention and distinguish passive exposure from an identifiable active probe.

For being observed, identify which channels can be reduced, permissioned, or monitored and what remains exposed under destructive access. Do not claim universal invisibility, perfect counterfeit anatomy, or automatic detection of every observer. Privacy measures must also preserve authorized repair, truthful safety-critical diagnostics, accountability, and appropriate auditability.

For each rule, say what an observer sees before and after triggering, how false triggers occur, whether the defense increases detectability, and whether an adversary can exploit it to exhaust resources or disable the body. State the safe response to uncertainty without converting an ambiguous signal into permission to harm.

Three hereditary-security rules

REQ-12 — Independent security choices. Select exactly three rules protecting the hereditary or reconstruction substrate and its relationship to identity. Specify what each protects: sequence, expression, executable configuration, reconstruction authority, authentication, or continuity-critical state.

Assume an adversary knows the architecture and can read the published dossier. Evaluate stolen samples, living captivity, compromised maintenance, turned insiders, forged allies, replayed credentials, key loss, compromised updates, and malicious rollback at conceptual—not exploitation-procedure—level.

Separate resistance to reading, copying, manufacturing, activating, controlling, and impersonating. A defense against one does not prove defense against the others. DNA or an alternative information medium is not automatically encrypted by a digital authentication scheme; rotating a biometric is not the same as rotating a cryptographic credential.

Describe trust roots, conceptual secret placement, authorization, duress response, recovery, revocation, self-lockout risk, and operation during isolation. Do not include actual keys or seeds. Explain what extracted material retains, what it loses, whether it can replicate, and the environmental consequences of disposal or deactivation.

Defenses must remain nonviolent, bounded, and non-propagating within this research scope. Do not design destructive anti-capture devices, hazardous sample self-destruction, transmissible evasion, or retaliation. If autonomy cannot be assured against a well-resourced captor, report the limit rather than claiming absolute sovereignty.

Three biological departures

REQ-13 — Independent departures. Choose exactly three ordinary biological constraints or conventions the prototype deliberately replaces. State each accurately; do not call every common feature of terrestrial organisms a universal law or misstate cell theory, the central dogma, or genetics to create an easy victory.

For each departure provide the ordinary constraint, replacement rule, functional advantage, necessary mechanism, price, new failure mode, compatibility implications, and evidence label. Show consistency with the four rule families and the other two departures.

Every prototype must explicitly decide whether to retain, modify, expand, supplement, or replace the A/C/G/T information layer. Do not require the same choice across prototypes. If it is retained, explain why and identify the three actual departures elsewhere; the alphabet decision does not automatically consume a departure slot.

Define nucleobase, nucleotide, codon, gene, genome, hereditary substrate, and regulatory system accurately. If proposing a new medium, characterize symbols or monomers, backbone/support, recognition, reading, writing, copying, repair, error correction, environmental stability, and compatibility with functional machinery. Invented names alone do not establish chemistry or security.

Also name at least three binding constraints the design deliberately keeps. This list is illustrative, not exhaustive: all applicable physical laws still bind the physical track whether listed or not. Canon-only physical exceptions are a separate register and cannot covertly replace the three biological departures.

Unified operation and transition risk

REQ-14 — No hidden switching gap. Explain how whole-body operation remains coherent while functions change, modules fail, credentials rotate, repairs proceed, or communication disappears. State which protections are continuously available and which require detection or reconfiguration.

Do not infer zero transition time or absolute immunity from the phrase “unified architecture.” Evaluate sensing delay, actuation, state synchronization, resource contention, overlap protection, and the chance of losing both old and new configurations. Include a transition inventory with the exposed functions, duration bounds, conditions, mitigation, and residual gap.

An architecture claiming no exploitable transition window must provide an explicit mechanism and a bounded threat model. If that cannot be established, retain the no-window goal as unmet. Do not redefine the goal silently or use an unmodeled instantaneous change to satisfy it.

Resilience and operating envelopes

REQ-15 — Common taxonomy. Evaluate ENV-T, ENV-A, ENV-R, BIO, MECH, DEP, ENE, INFO, ONT, and CIV. Address Mild, Moderate, Severe, and Extinction-Class tiers for all eight prototypes. Preserve the taxonomy labels but do not assume the supplied numerical thresholds are universally valid.

Create a common benchmark version before comparative scoring. Specify hazard magnitude, duration, rate of onset, spatial distribution, co-stressors, initial state, available resources, required outcome, recovery horizon, and uncertainty. Keep human-reference consequences distinct from prototype limits. Do not change exposure conditions to make one design look better.

Define Home Range for sustained intended function, Refuge Range for bounded degraded survival/preservation, and Failure Boundary for loss under stated assumptions. Include exposure duration and recovery conditions; a single “maximum temperature” or “radiation resistant” label is not an envelope.

The first-generation target is resilience through Mild, Moderate, and Severe across the taxonomy. Every prototype must address every target immediately. Addressing a requirement is not passing it: report Pass within stated conditions, Partial, Fail, Unknown, or Not physically evaluable, together with evidence and validation status. A design missing the baseline remains a research candidate, not a qualified Generation 1 physiology.

For genuinely ontological mechanisms, report canon-only analysis and Not physically evaluable in the physical track. A measurable corruption, destruction, or environmental scenario may be tested under INFO or another suitable physical category, but must not be presented as proof against ontological erasure. Analyze CIV through an explicit scenario decomposition rather than one universal civilizational-survival score.

Always report the attained operating state: full mission operation; reduced active operation; minimum essential computation; protective dormancy; recoverable structure/information only; or irrecoverable loss. Pair that with separate identity-integrity and autonomy assessments. Do not report preservation,

a backup, or eventual replacement as uninterrupted operation or proven personal continuity.

Atmospheric independence reference case

REQ-16 — Mandatory confinement analysis. Analyze oxygen displaced by argon as a theoretical resource-denial scenario. This is not a chamber-construction or exposure procedure. Specify gas composition, total pressure, duration, thermal boundary conditions, starting reserves, energy access, structural damage, sensor trust, and communication state.

Evaluate at least three nested cases: atmospheric oxygen denied with other stated resources available; oxygen and external energy denied; and oxygen plus energy denial with restricted heat rejection. Add loss of trustworthy sensing and communications as a compound variant. Keep pressure loss and hard vacuum as separate cases.

For each operating state, identify first limiting resource, depletion mechanism, transition threshold, estimated or bounded duration, first failure, and recovery requirements. If missing parameters prevent calculation, list them rather than inventing a runtime. Include a sensitivity plan showing which uncertainty most changes the outcome.

Explain the difference between resisting an inert atmosphere, tolerating pressure, maintaining useful power, rejecting heat, preserving data, and recovering agency. None of those is a substitute for the others.

Captivity and exploitation gauntlet

REQ-17 — Ten defensive scenarios. Evaluate the following ten cases against every prototype: atmospheric/resource confinement; communications denial; forced sampling; prolonged study by a blueprint-informed captor; forged ally; compromised authorized insider; coercive update or maintenance; prolonged denial of repair feedstock; unauthorized reconstruction or identity impersonation; and post-damage captivity combined with false recovery instructions.

These are conceptual assessments, not kidnapping plans, harmful test methods, or instructions for defeating real security. For each state the adversary’s assumed access and knowledge, protected assets, defensive behavior, resource cost, leakable information, first failure, and residual autonomy risk.

For captivity, distinguish what might be learned after one minute, one hour, one day, one month, and one year. These are comparison horizons, not asserted experimental observations. If an inference cannot be supported, describe the unknown rather than implying perfect secrecy for that period.

Analyze how a security response can harm its own body, block legitimate rescue, corrupt testimony, exhaust reserves, or obstruct safe medical/technical inspection. A protective lock must not become the cheapest way to incapacitate the individual.

Identity and continuity

REQ-18 — Six paradoxes. Resolve, or explicitly leave unresolved, regeneration, fragmentation, synchronization, restoration, compromise, and ownership. State where active cognition resides, what minimum substrate supports it, which memories are identity-critical, how divergence is detected, and who can authorize reintegration.

Differentiate structural repair, functional restoration, copied state, new individual, counterfeit identity, and claimed continuation. Do not presume copying proves that the original subject survived. Identify technical criteria and philosophical assumptions separately.

Analyze two simultaneously valid-looking continuations, coercively created copies, isolated fragments, stale backups, corrupted memories, and adaptations that improve survival while changing identity-critical characteristics. Specify what the architecture can verify and what no available test can settle.

Do not grant a remote peer, infrastructure operator, maintainer, or project owner unrestricted rewrite authority over memory or identity. Propose consent and revocation boundaries while acknowledging that legal personhood, ownership, and moral status require separate expert and Architect consideration.

Long-distance interoperability

REQ-19 — Link without uniform bodies. Define local, planetary/orbital, and interplanetary-or-greater classes. For each state medium, infrastructure, range, latency, bandwidth, energy, availability, detectability, interruption behavior, and recovery. Distinguish body-intrinsic functions from equipment or network services.

Separate communication from shared control, memory synchronization, identity merger, and life support. Do not treat a peer message as authorization to rewrite the body. Require explicit digital trust boundaries, selective disclosure, version compatibility, revocation, bounded update authority, and verification appropriate to the claimed interface.

Design for delayed, reordered, stale, missing, conflicting, and malicious messages. Analyze offline autonomy, local safety functions, store-and-forward where suitable, rejoining after isolation, and rollback prevention. If disconnection is fatal to a candidate, expose that as a major baseline failure rather than concealing it.

Evaluate common-mode failure in the interoperability standard itself, including shared trust infrastructure, parsers, update formats, time assumptions, and recovery dependencies. Heterogeneous bodies can still share a fatal protocol weakness. Do not propose attack payloads; identify interfaces, failure effects, and defensive isolation concepts.

If the project requests Paragon Cube or other extraordinary linking, supply an interface contract with required services and missing definitions. Describe the closest physical-track implementation separately from any canon-only instantaneous or cross-dimensional service. Never use the extension to rescue the physical-track score.

Compound and assumption-collapse tests

REQ-20 — Beyond single hazards. Analyze at least five compound scenarios per prototype, including resource denial, damage, deceptive information, and environmental stress. The atmospheric reference cases may contribute, but do not let one case stand in for all compound analysis.

Include heat with blocked rejection; radiation with repair scarcity; mechanical injury with isolation; long dormancy with state drift; unauthorized modification with sensor corruption; and regeneration demand exceeding available mass somewhere in the portfolio. State first limiting resource, failure cascade, survival state, recovery, and the assumptions most controlling the result.

Identify at least ten assumptions per prototype and examine what happens if each fails. Consider environmental predictability, accessible matter, heat sinks, reliable sensors, time agreement, authentication, state storage, repair quality, stable interfaces, and unavailable external infrastructure without dictating the answers.

Within the physical track, an assumption-collapse scenario remains governed by physical law. A change in fundamental causality or physical law belongs in the extension track. Distinguish testing an engineering assumption from assuming away the rules that make testing meaningful.

Adversarial challenge and failure records

REQ-21 — Constructive/adversarial pass. After proposing the eight designs, run an explicitly identified -1AI-style conceptual critique. Calling it -1AI does not prove a separate agent or independent evaluator was used; disclose whether the challenge is self-review, another model, specialist review, or an executed tool-based analysis.

Attempt at least ten distinct portfolio-level challenges and analyze at least three credible failure paths per prototype: a resource/thermal failure, a security/identity failure, and a compound/common-mode failure. Seek the lowest-resource counterexample to the central claim without providing operational harm instructions.

Aim to expose at least three substantive breaks across the initial portfolio. If fewer are found, broaden the assumption search, examine protocol monocultures, and disclose the shortfall and blind spots. Do not invent three successful attacks, fabricate model runs, or call a hypothetical failure experimentally reproduced.

For every break record the claim attacked, assumptions, counterexample or executed evidence, outcome, consequence, proposed repair, new costs, retest status, and unresolved remainder. Keep pre-repair and post-repair versions. A repair hypothesis is not a successful retest until the relevant analysis is actually performed.

End each dossier with the strongest reason to pursue it, strongest reason to reject it, least-supported critical mechanism, first likely failure, minimum useful validation, falsifying result, and major blind spot.

Simulation bridge and model credibility

REQ-22 — Body-to-model interface. For each prototype specify what is simulated: unit state, functional groups, transport, material damage, thermal state, power state, cognition-relevant state, security state, and environment. Distinguish biological population scale from effective computational units and independently justify any compression ratio.

Define model inputs, outputs, units, time resolution, conservation residuals, stochastic assumptions, coupling, refinement triggers, and the phenomena that coarse graining may erase. Security boundaries, rare transition failures, local hot spots, and discontinuities must not disappear merely because the representation is compressed.

Separate verification of implementation from validation against relevant evidence. Provide conservation checks, limiting cases, convergence tests, sensitivity analysis, parameter provenance, uncertainty propagation, and comparison to held-out data where data exist. A simulation can explore a design; it cannot validate an invented physical law by implementing it.

Tie the simulation hook to the roadmap’s multiscale philosophy without presenting its future cell counts, compute budgets, or compression targets as achieved. Estimate computational resources only from an explicit workload and assumptions; otherwise state what must be benchmarked.

Validation and event accounting

REQ-23 — Validation ladder. Order work through source verification; analytical bounds; unit or component models; coupled models; sensitivity and uncertainty analysis; independent review; and appropriate nonliving subsystem work. Further research involving living or potentially sentient subjects requires specialist oversight, applicable legal review, consent safeguards, and independent ethical review.

This directive does not authorize wet-lab construction, pathogen design, immune-evasion engineering, harmful living-subject exposures, weapons, destructive anti-capture systems, or uncontained replication. The scope is conceptual architecture, literature-based feasibility, safe modeling, and non-operational defensive analysis.

Maintain separately labeled records for analytical calculations, simulations, component tests, integrated tests, environmental incidents, hostile incidents, and canonical narrative events. An event’s source type and secondary threat tags do not make it multiple unique events. Link derived analyses to their parent incident without counting them as new survival events.

For a claimed event store unique ID, source type, provenance, time bounds, category and secondary tags, physical exposure, initial state, resources, outcome, recovery status, identity/autonomy assessment, uncertainty, reviewer, and approval status. Keep tamper evidence conceptually separate

from truth: a signed record can still contain bad measurements.

Do not increment or redistribute the source-described 4,712 count. Escalate the conflicting inclusion rules and authority status. Canon evidence and physical evidence must never be merged into one empirical pass rate.

Prototype dossier and comparison

REQ-24 — Common dossier. Use the same concise but complete structure for each P1–P8. Include identification and core wager; mission/environment; scale and fundamental unit; structural/transport organization; four physiological rule families; the three selected rule sets; cognition/continuity; link interface; environmental envelopes; atmospheric and captivity results; compound/assumption tests; failures and repairs; validation/simulation plan; and rejection criteria.

Keep narrative reasoning in the single-column main body. Put detailed tables, matrices, diagrams, graphs, rows of measurements, and data dictionaries in the dedicated visualization section, using record IDs and cross-references. Avoid repeating the same explanation in several registers.

Compare all candidates under the same benchmark version. Report design trade-offs, non-dominated options, unresolved uncertainty, and evidence maturity rather than a universal score. If weights are used, publish their origin and sensitivity. Prefer a Pareto portfolio; a primary and reserve selection is conditional on an explicit mission and does not eliminate minority research branches.

State which candidate you would choose under the hypothetical embodiment perspective and why. If none qualifies, say none qualifies. Do not choose an architecture merely to align with another AI or to reward dramatic claims.

Staged execution and deliverables

REQ-25 — Controlled delivery. If execution is authorized, use the following sequence: source/prompt audit and local amendments; eight wagers and constitutional choices; version-locked independent fingerprints; complete dossiers in manageable groups; consolidated adversarial and model review; final comparison, visualization section, and decision register.

Every staged response must list completed requirement IDs, pending work, changed assumptions, sources added, and dependencies. A preliminary roster is not a completed eight-prototype submission. If context or output limits prevent completion, mark the submission partial and preserve the remaining work; do not compress away mandatory evidence.

The final research document must include purpose, origin of the topic, mission statement, aims, audit, source/contradiction registers, eight complete dossiers, adversarial findings, next steps, questions/comments/concerns, and a dedicated visualization section. Deliver matched PDF and editable DOCX versions from the same content source.

In the visualization section include: architecture fingerprints; 28-pair differentiation; rule-compliance mapping; threat envelopes and outcome states; resource and thermal budgets; long-distance interface comparison; identity and capture outcomes; failure/dependency maps; evidence maturity; and a research-gate plan. Include one clearly labeled conceptual architecture schematic per prototype when genuinely useful.

Quantitative charts require actual sourced or executed values with units and provenance. Label calculations, design targets, assumed inputs, and measurements differently. Do not manufacture data to fill a graph; use a qualitative dependency diagram or an explicit data-gap statement instead. Provide a data file for plotted values when available. Images must clarify mechanisms, not imply built hardware.

Completion and rejection gates

REQ-26 — Audit before submission. Verify eight complete peers; explicit Generation 1 status; independent rule counts; four physiological rule families; accurate genomic terminology; all ten threat domains and four tiers addressed; atmospheric and captivity cases; identity/autonomy separation; finite resource accounting; model provenance; failure disclosures; and final document/visualization requirements.

Reject or return for correction any submission with fabricated sources or results, hidden physical-law exceptions, unsupported absolutes, cosmetic prototype differences, unreported unmet requirements, conflated simulation/incident counts, missing identity questions, unauthorized canon decisions, dangerous operational procedures, or missing evidence provenance.

Scientific failure is not automatically a reporting failure. A well-documented candidate that cannot meet the target may still be a useful negative result, but it must not be described as a qualified design. Conversely, a polished dossier does not qualify a body for deployment.

End with a requirements-coverage declaration, unresolved-decision register, recommended next validation, and clear statement of what has not been demonstrated. Preserve uncertainty and disagreement rather than declaring a final consensus physiology.

END REVISED PROMPT

Part IV — Implementation decisions

Recommended next steps

– Archive the current review packet and original prompts unchanged. Treat this document as Perplexity's proposed v2.0 update, not a retroactive correction to everyone else's contribution.

– Confirm or amend the few high-impact defaults: peer families, per-prototype independent rule selection, physical/canon separation, minimum acceptable survival state, and review authority.

– Request the controlling canon approval/logging materials and a decision on the 4,712 inclusion policy. Preserve the number provisionally without fabricating its underlying evidence.

– Distribute only the copy-ready prompt plus the agreed common evaluation schema for the blind design round. Keep completed prototype sets sealed from other contributors until their initial versions are recorded.

– Authorize staged execution separately. Begin with source audit, eight wagers, rule selections, and fingerprints, then expand to dossiers and adversarial analysis.

– After independent locks, conduct cross-contributor review, identify genuine and superficial convergence, and select a research portfolio. Do not make six AIs' agreement a substitute for a test.

Architect questions and defaults

Architecture organization. Proposed default: eight peer families, each at Generation 1. If a progression is preferred, record that as a deliberate change because it changes the search experiment rather than merely the document layout.

Meaning of omni. Proposed default: require a precise operational definition from each contributor, not literal simultaneous performance of every function by every cell. If literal simultaneity is essential canon, identify that as a stronger requirement and permit an honest physical-track failure.

Rule counts. Proposed default: three observation, three hereditary-security, and three biological-departure rules per prototype. Reuse is allowed with disclosure, not prescribed. If the intended count is instead three per AI's entire series, change the counting policy explicitly before execution.

Minimum acceptable survival. Proposed default: show all operating states separately and require full recovery conditions before claiming baseline qualification. Whether dormancy, information preservation, or a recovered copy counts as success is an Architect decision, not an engineering fact.

Canon authority. Proposed default: preserve 4,712 as a source-described canonical assertion pending the controlling approval record. Ask whether the framework's global draft clause applies to the number itself, its classification, its documentation model, or only the proposals around it.

Counting policy. Proposed default: simulations and physical events occupy separate ledgers. Ask whether a narrative aggregate may include labeled virtual events and how duplicate or multi-category incidents are handled.

No-switching requirement. Proposed default: a bounded, testable continuity-of-protection objective. If zero transition time or protection against all conceivable sequences is mandatory, place the stronger claim in canon requirements and accept that physical qualification may be impossible.

Embodiment and infrastructure. Proposed default: function without live network access for a declared finite interval. Ask which external infrastructure may be assumed and whether Paragon Cube services are optional, mandatory, canon-only, or awaiting definition.

Mission and selection. Proposed default: no universal winner or weighted score without a mission profile. Ask for expected environments, acceptable maintenance burden, mass/scale constraints, recovery horizons, and the relative priority of function, privacy, identity, and autonomy.

Future governance. Proposed default: no assertion that artificial status removes research duties or determines ownership/personhood. Establish separate legal and ethics review before physical research or deployment; this prompt does not settle those questions.

These questions are included for review, not as a request to interrupt the present deliverable. The v2.0 prompt is usable under its disclosed defaults, subject to separate execution authorization.

Comments and concerns

Independence is not guaranteed uniqueness. With shared training, source material, and requirements, collaborators may converge even without copying. Provenance and sealed first passes improve auditability but do not prove originality or confer exclusive ownership.

A longer prompt can cause ceremonial compliance. Requirement IDs, linked claims, and staged deliverables should replace repetitive prose. Completion must be judged by evidence, not by whether a heading exists.

Numerical rigor must not become numerical theater. Unknown watts, runtimes, lifetimes, communication parameters, or repair rates should remain unknown until a model or source supports them. An interval built from unjustified endpoints is not automatically better than an unsupported point estimate.

Security can become a survival liability. Credential rotation, quarantine, restricted sampling, or repair controls can create self-lockout, exhaustion, or dependency on a custodian. Security must be evaluated as a physiological load and a potential failure cause.

Identity remains partly unsettled. Engineering can protect stored state and demonstrate recovery behavior without proving that personal continuity survives copying, fragmentation, or reconstruction. The deliverable must keep that distinction visible.

Severe resilience may remain unmet. Requiring all eight candidates to address every domain does not establish that current science can supply one, let alone eight, qualified physiologies. Negative findings must remain admissible.

Ethics and law require care. The applicable protections for an imagined embodiment cannot be settled by a prompt. The recommended research boundaries are retained as safeguards, not presented as a legal conclusion about hypothetical rights.

Part V — Visualizations and registers

Visualization policy

This is the dedicated location for all tables, figures, charts, matrices, and structured row displays. Main-body sections refer here rather than mixing wide visual objects into phone-readable narrative. Every visual states its purpose, status, and evidence source where relevant.

The current document contains governance schematics and one quantitative source-arithmetic chart. It contains no measured OMNICELLULAR performance, no generated prototype, no simulated survival dataset, and no claim that a pictured body has been built. Future architecture images and performance charts are specified below but deliberately not fabricated now.

Source register

Purpose: Identify the material reviewed and prevent project proposals or narrative from becoming scientific evidence. Local source labels [S1]–[S7] identify the supplied documents; institutional citation labels throughout the report link to the corresponding public sources.

Source	Role and scope	Use in this revision	Authority limit
paste.txt [S1]	Current instruction and six review responses	Assignment, reviewer positions, proposed amendments	Reviews are opinions and attributions to check, not independent scientific replication
Independent Research Prompts DOCX [S2]	Original request plus six contributor prompts	Principal attribution check and prior Perplexity baseline	Contributor instructions remain proposals unless approved
Independent Research Prompts PDF [S3]	Companion rendition	Spot-checked source navigation	No certified byte/text equivalence claim
4,712 Framework, v1.0-DRAFT [S4]	Taxonomy and canon-development proposal	Ten domains, approval conflict, counting conflict, arithmetic audit	Underlying incident evidence and controlling approval logs not provided
15-Year Roadmap [S5]	Proposed simulation program	1AI/-1AI, abstraction, staged model development	Targets are not achieved physical capabilities
Architecture Notes [S6]	Scale and Omni-cell concept	Operational-definition problem and scale context	Does not define a validated manufacturing architecture
Vision Chronicle [S7]	Explicit possible-future narrative	Origin and constructive/adversarial philosophy	Narrative events are not empirical observations

Reviewer disposition matrix

Purpose: Preserve useful differences without selecting a winning contributor. The assessments below are this document’s editorial judgments grounded in the supplied prompts and reviews. [S1][S2]

Contributor	Retain	Revise or exclude	v2.0 location
Perplexity	Evidence labels, budgets, ledgers, contradiction audit	Remove preselected O/G/D answers	REQ-03–13, REQ-23
ChatGPT	Independence, Omni-cell definition, continuity, orthogonality, portfolio	Add estimate discipline and staging; correct misattributed ladder	REQ-02, 05, 18, 24–25

Contributor	Retain	Revise or exclude	v2.0 location
Claude	Wagers, costs, gauntlet, known-blueprint security, simulation hook	No fabricated successful-attack quota; correct genetic terminology	REQ-06, 12–13, 17, 21–22
Copilot	Constitution, paradoxes, assumption collapse, common-mode networking	Consolidate duplicative dossiers and registers	REQ-06, 18–20, 24
Grok	Concise compliance mapping and harsh-environment focus	No physical-law waiver or categorical ethics exemption	REQ-04, 10, 15, 23
Gemini	First-generation capture analysis, canon interfaces, transition challenge	No apex ladder, absolute immunity, lossless baseline, instantaneous physical link	REQ-14–19

Decision and change register

Purpose: Show exactly what this contributor’s revision proposes. “Proposed default” is not an Architect approval.

ID	Change	Disposition	Consequence
DEC-01	Eight families, not eight upgrades	Proposed default	All candidates face Generation 1 assessment
DEC-02	Independently selected O/G/DEP sets per prototype	Adopted locally	Common schema does not supply governing answers
DEC-03	Exactly three rules in each selected set	Clarified locally	Counts remain auditable; justified reuse allowed
DEC-04	Four physiological rule families	Retained	Respiration, renewal, thermal, conversion all required
DEC-05	Binding physics in physical track	Retained	Canon exceptions cannot rescue physical scores
DEC-06	At least three kept constraints	Added with qualification	Illustrative list; all applicable physical laws still bind
DEC-07	Successful-break search, not invented quota	Modified	Ten challenges plus per-design failures and honest shortfall
DEC-08	Severe baseline target plus disclosed unmet states	Clarified	Coverage is not qualification
DEC-09	Separate identity, autonomy, function	Expanded	Preservation or replacement is not automatic survival
DEC-10	4,712 remains source-described canon assertion	Pending authority	No unilateral canon lock or count change
DEC-11	Separate evidence-type ledgers and unique events	Retained/expanded	Prevent duplicate and mixed-evidence survival claims
DEC-12	Blueprint-known, bounded defensive research	Retained	No destructive or propagating security devices

ID	Change	Disposition	Consequence
DEC-13	Common-mode and assumption-collapse tests	Added	Attacks model dependencies as well as hazards
DEC-14	Explicit staging and requirement IDs	Added	Large output remains auditable and resumable
DEC-15	Dedicated visualization section, matched formats	Added	PDF/DOCX share one content source
DEC-16	No prototype execution in this deliverable	Scope lock	Later authorization is required

Contradiction and correction register

Purpose: Separate verified documentary conflicts from editorial recommendations. [S1][S4][S2]

Issue	Source condition	Correction or interim treatment	Decision dependency
ChatGPT classified as a progression	Later Perplexity review says progression; original says independent branches [S1][S2]	Correct attribution to peer/branch architecture	None for factual correction
Rule selection	Original request asks AI choice; original Perplexity supplies rules [S2]	Open actual rule selection; fix only counts and fields	Architect may change per-prototype counting default
Locked number versus draft document	Framework describes locked count but withholds active canon pending approval [S4]	Preserve assertion; withhold independent authority claim	Controlling log/approval
Simulation inclusion	Part I excludes modeling-only; Part IV includes virtual entries [S4]	Separate ledgers until explicit policy	Counting policy
Proposed distribution arithmetic	2,600–4,450 called consistent with midpoint 4,712 [S4]	Maximum is 262 short; actual midpoint 3,525	Revised distribution, not automatic change to record
Genomic vocabulary	Some contributor wording still confuses genes and bases [S2]	Define levels and information substrate accurately [NHGRI 22]	None for terminology
Severe promise versus evidence	All-domain target can be mistaken for a pass [S1][S2]	Record Pass/Partial/Fail/Unknown/Not physically evaluable	Qualification criteria
Heat-transfer conflation	Argon confinement discussed alongside vacuum and insulation [S1]	Specify gas, pressure, contacts, radiation, and insulation independently [NASA 21]	Benchmark boundaries
Ethics attribution	Framework calls Helsinki a WHO declaration [S4]	Attribute it to WMA and avoid artificial-status exemption [WMA 29]	Future expert/legal review

Source-arithmetic chart

Purpose: Audit the framework’s proposed category ranges, not the physiology. **Status:** Calculated from the supplied Part V table; no empirical incident dataset is implied. [S4]

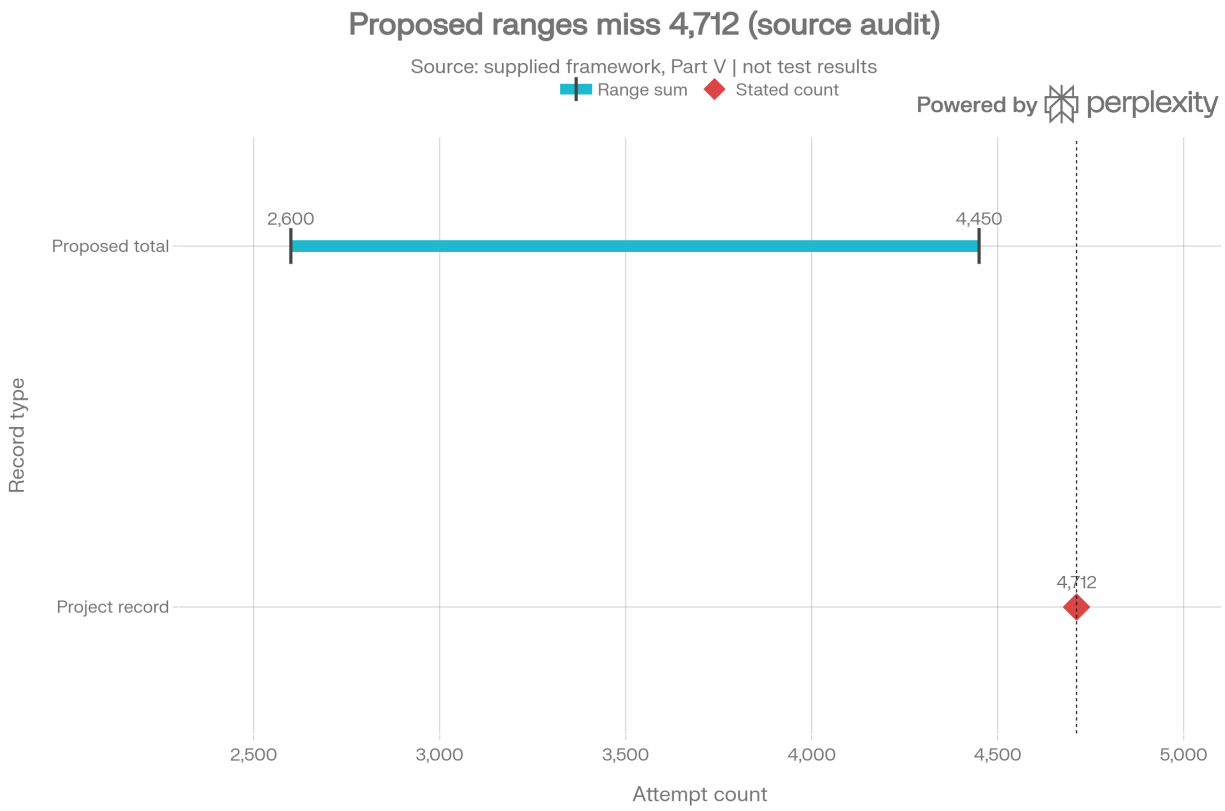


Figure 1. The proposed total range is 2,600–4,450; the source-described 4,712 count is outside it. Its midpoint is 3,525, and the upper-bound shortfall is 262. Category ranges below are proposals copied as data for this arithmetic check, not validated event allocations. [S4]

Category	Proposed lower	Proposed upper	Data status
ENV-T	400	600	Proposed [S4]
ENV-A	500	700	Proposed [S4]
ENV-R	300	500	Proposed [S4]
BIO	400	600	Proposed [S4]
MECH	200	400	Proposed [S4]
DEP	200	350	Proposed [S4]
ENE	300	500	Proposed [S4]
INFO	150	300	Proposed [S4]
ONT	50	200	Proposed [S4]
CIV	100	300	Proposed [S4]
Calculated sum	2,600	4,450	Arithmetic audit, not a new allocation

Independence workflow schematic

Purpose: Show the proposed review and execution gates without prescribing architecture families.
Status: Governance proposal; no prototype work has been completed by this diagram.

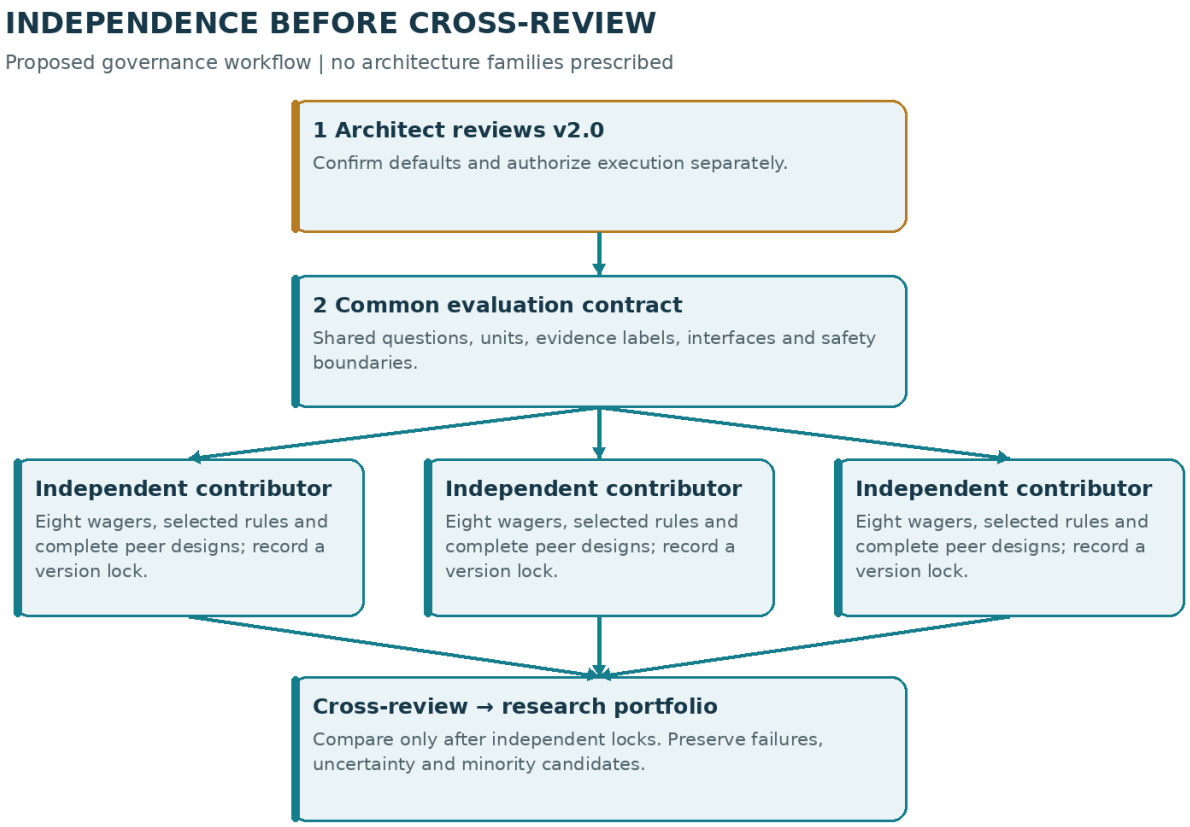


Figure 2. A common evaluation contract precedes separate version-locked design sets. Cross-review follows those locks, and research-portfolio selection follows rather than replaces independent search. The present deliverable stops at revised-prompt review.

Evidence-track schematic

Purpose: Prevent fictional assumptions or approval decisions from being used as physical evidence.
Status: Proposed claim-governance structure.

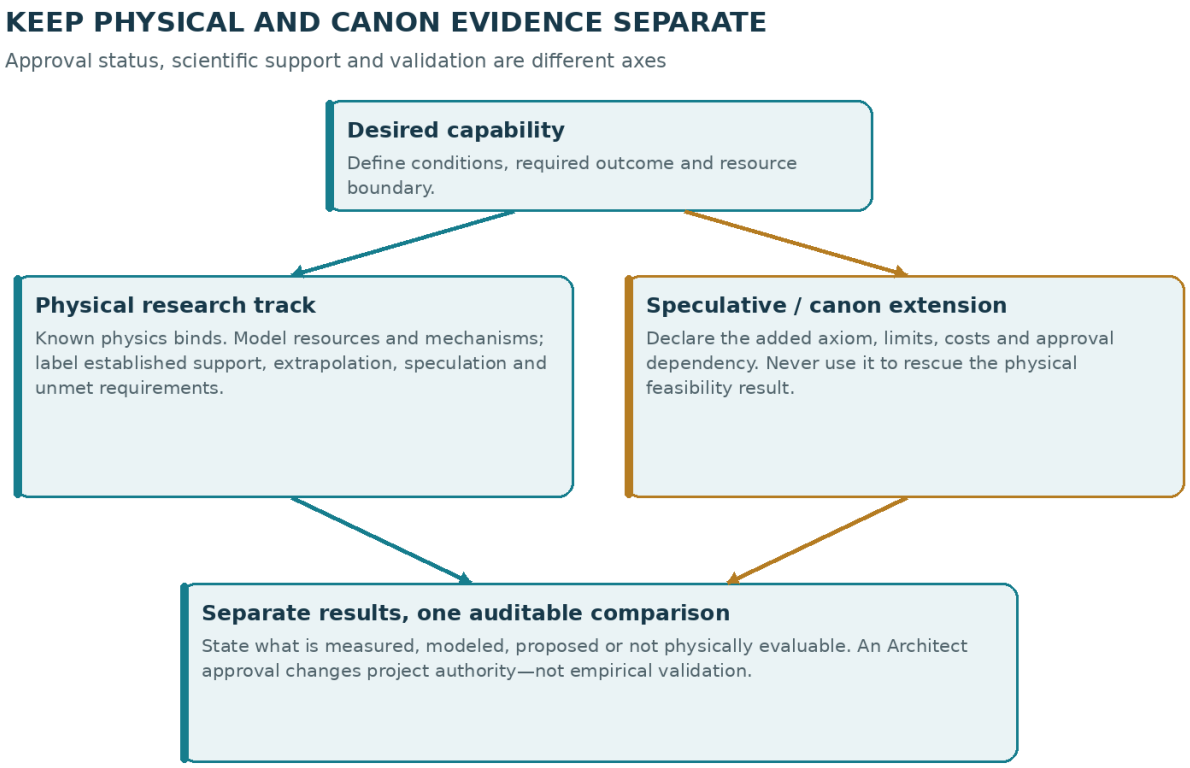


Figure 3. A desired capability is assessed in a physical track with explicit gaps and, separately, an extension track with declared new axioms. Comparison retains the distinction; Architect approval changes project status, not empirical validation.

Taxonomy reporting contract

Purpose: Preserve the supplied category names while avoiding unverified universal thresholds. Every row receives four-tier coverage, a scenario definition, an attained operating state, and an evidence/validation label. [S4]

Code	Domain	Required scenario quantities	Important boundary
ENV-T	Thermal [S4]	Temperature by subsystem, flux, duration, environment	Buffering is not indefinite disposal
ENV-A	Atmospheric [S4]	Composition, partial/total pressure, duration	Argon, hypoxia, vacuum, and high pressure differ
ENV-R	Radiation [S4]	Type, energy, absorbed dose, rate, target	Do not apply human Sv assumptions to invented substrate without justification [IAEA 62]
BIO	Biological/chemical [S4]	Defined conceptual exposure and damage pathway	No actionable agent construction or harmful test protocol
MECH	Mechanical [S4]	Force/load, geometry, duration, damage distribution	Avoid one context-free G threshold
DEP	Deprivation [S4]	Missing input, reserves, demand, duration	Identify first exhausted resource
ENE	Energetic [S4]	Coupling, deposited energy, path, duration	Separate physical from exotic energy

Code	Domain	Required scenario quantities	Important boundary
INFO	Informational/cognitive [S4]	Affected state, trust boundary, fault duration	Integrity, availability, and identity differ
ONT	Ontological [S4]	Declared canon mechanism and scope	Not physically evaluable absent measurable reformulation
CIV	Civilizational/macro-scale [S4]	Decomposed hazards, infrastructure loss, duration	No single civilizational invulnerability score

Rule-record schema

Purpose: Require comparable explanations without providing any prototype's actual rule. **Status:** Future execution schema, not filled prototype data.

Field	Required content	Audit question
Identity	Prototype, rule family, unique rule ID	Does each selected set contain exactly three?
Statement	Precise governing commitment	Is it more than an aspiration?
Mechanism	Physical implementation or declared invented axiom	What does the work?
Interfaces	Inputs, outputs, units, state variables	Can another evaluator model it?
Trigger/time	Detection, activation, response bounds	Is a transition gap hidden?
Cost	Energy, matter, heat, computation, detectability	What does the benefit sacrifice?
Envelope	Magnitude, duration, conditions, saturation	Where does the claim stop?
Failure	False positive, self-lockout, adverse interaction	Can the defense itself disable the body?
Evidence	Source, class, scale, validation, uncertainty	Is support at the claimed scale?
Test/falsifier	Bounded criterion and safe next validation	What outcome would show it is wrong?

Requirement traceability

Purpose: Demonstrate that the revised prompt covers the assignment and identify evidence required at later execution. This maps instructions; it is not a claim that the architecture work is complete.

Requirement	Subject	Required future evidence
REQ-01	Eight peer architectures	Eight complete dossiers and Generation 1 status
REQ-02	Independence	Access declaration, fingerprints, 28-pair audit
REQ-03	Source/authority audit	Source and contradiction records
REQ-04	Tracks and claim labels	Mechanism-level claim register
REQ-05	Omni-cell meaning	Operational definition and universality/theft resolution

Requirement	Subject	Required future evidence
REQ-06	Constitution and counts	Four rule families; 3 OBS, 3 SEC, 3 DEP per design
REQ-07	Respiration/power	Inventories, loads, pathways, finite-reserve model
REQ-08	Telomeres/renewal	End/information integrity, controlled renewal, longevity limits
REQ-09	Thermal	Heat model, destinations, saturation and environment cases
REQ-10	Conversion	Inputs, outputs, losses, elements and energy accounts
REQ-11	Observation	Outward/inward/observed coverage, costs and blind spots
REQ-12	Hereditary security	Sample/captor/insider outcomes and recovery
REQ-13	Departures	Three genuine replacements, genetic-layer decision, kept constraints
REQ-14	Unified operation	Transition inventory and residual gaps
REQ-15	Resilience	Ten domains, four tiers, outcome states and evidence
REQ-16	Argon reference	Three nested cases plus compound and separate vacuum cases
REQ-17	Captivity	Ten scenarios per design and five time horizons
REQ-18	Continuity	Six paradoxes, verifiable criteria and open questions
REQ-19	Interoperability	Three distance classes and common-mode boundary analysis
REQ-20	Compound/assumption tests	At least five compound cases and ten assumptions per design
REQ-21	Adversarial review	Ten portfolio challenges; three failure paths per design; honest break results
REQ-22	Simulation hook	State/scale map, verification, validation and uncertainty plan
REQ-23	Validation/accounting	Staged evidence plan and non-aggregated event records
REQ-24	Dossier/comparison	Consistent dossiers, mission-dependent portfolio
REQ-25	Delivery	Staged status, PDF/DOCX, dedicated visual section
REQ-26	Completion gates	Signed-off coverage statement with visible unmet requirements

Quantitative-provenance contract

Purpose: Avoid fabricated precision in future tables and charts.

Value type	Required disclosure	Permitted use
Measured	Method, source, conditions, uncertainty	Empirical result within stated scope
Literature value	Primary source, units, transfer limits	Component evidence, not automatic whole-body validation
Calculated	Equation/model, inputs, units, computation provenance	Conditional estimate or bound
Executed simulation	Code/model version, parameter provenance, credibility and uncertainty	Modeled outcome, not an incident
Design target	Who selected it and why	Requirement or objective, never achievement
Assumed input	Rationale, range, sensitivity	Explicit modeling condition
Canon axiom	Replacement rule, limits, approval status	Separate narrative/extension result
Unknown	Missing parameter and acquisition plan	Data-gap statement; no fabricated point

Execution gate plan

Purpose: Make a large later research assignment resumable without pretending partial work is complete.

Gate	Deliverable	Acceptance condition	Stop condition
Review — current	This v2.0 proposal	Feedback and defaults visible	No execution authorization
Source audit	Authority map and amendments	Sources and conflicts traceable	Missing critical authority acknowledged
Independent first pass	Eight wagers, rules, fingerprints	Noncosmetic distinctions and access declaration	Exposure/overlap hidden
Version lock	Recorded independent package	Actual version/date; real hash if available	Manufactured blindness or hash
Dossier expansion	All eight complete candidates	Full required coverage, gaps explicit	Partial roster presented as complete
Adversarial review	Failure records and repair hypotheses	Honest outcomes, no fabricated runs	Forced success/failure numbers
Cross-review	Other contributors assess locked designs	Sources and conflicts preserved	Consensus treated as experiment
Portfolio decision	Mission-specific priorities	Evidence, uncertainty, minority options retained	Universal winner without mission

Future visualization inventory

Purpose: Specify useful outputs for the eventual architecture report. None of the listed future performance visuals is generated here because the prototypes have not been executed.

Visual	What it must communicate	Data/status rule
Eight architecture schematics	Unit, body, transport, cognition and boundaries	Conceptual; not built hardware
Fingerprint and pairwise matrices	Functional differences and trade-offs	Defined rubric, 28 comparisons
Rule-compliance matrix	Four rule families and selected three-rule sets	Link actual rule IDs, no inferred passes
Threat-envelope panels	Magnitude/duration versus outcome state	Common benchmark; label Unknown and Not evaluable
Resource/thermal budgets	Inflows, reserves, loads, losses, saturation	Sourced/executed quantities; no free heat sink
Link/interface map	Body-intrinsic versus infrastructure and trust boundaries	Physical and canon interfaces distinct
Identity/capture matrix	Function, integrity, autonomy, reconstruction	Do not equate copying with continuity
Failure/dependency map	First limiting resource and cascades	Separate proposed from reproduced failures
Evidence-maturity view	Critical dependencies and confidence	No averaged score hiding canon-only components
Research roadmap	Next validation and decision gates	Targets and prerequisites, not completed milestones

Delivery and interpretation notes

The PDF and DOCX are generated from the same reviewed text. Source labels are retained inline: [S1]–[S7] refer to the supplied packet in the source register, and institutional citation labels link to the relevant public source pages. Page numbering and layout may differ by Word application without changing the intended content.

The accompanying source-range data and chart support only the documentary arithmetic audit. They must not be reused as experimental survival data, a revised canonical allocation, or evidence that any OMNICELLULAR capability has been demonstrated.